INCH-POUND
MIL-DTL-22442/41B(AS)
15 September 2003
SUPERSEDING
MIL-DTL-22442/41A(AS)
5 January 1998

DETAIL SPECIFICATION SHEET

CABLE ASSEMBLY, AIRCRAFT AUDIO, ADAPTER, SURVIVAL RADIO

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-22442.

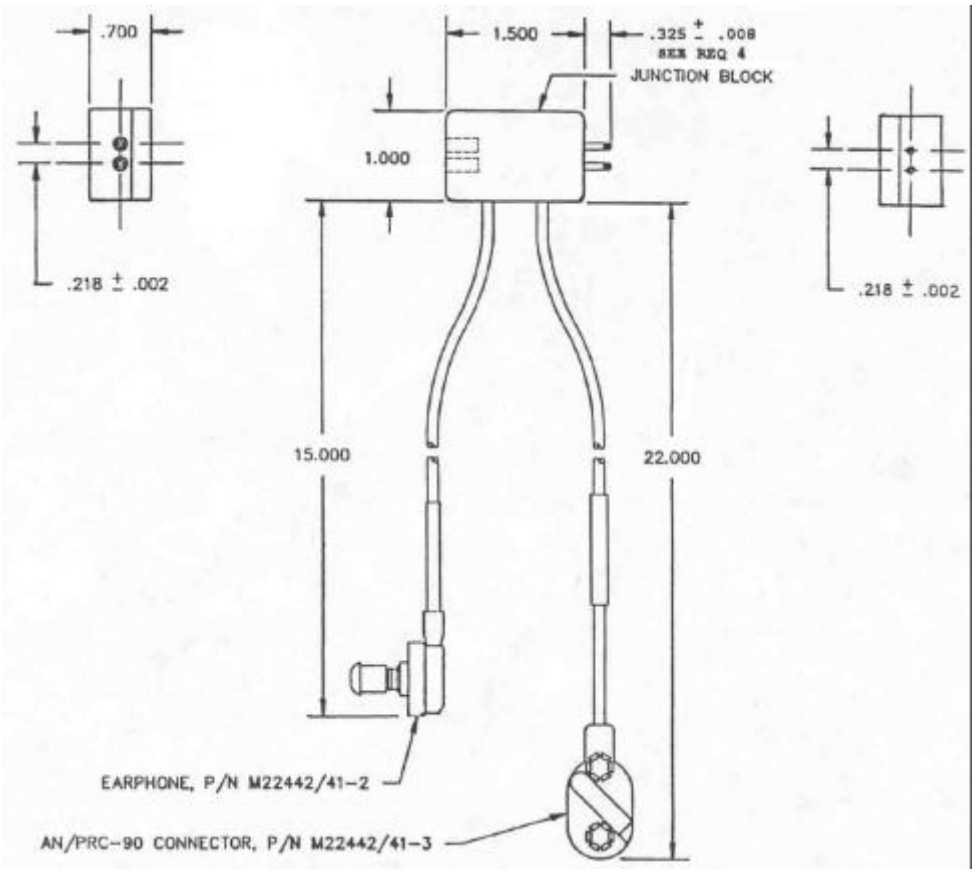


*FIGURE l. Cable assembly P/N M22442/41-l.

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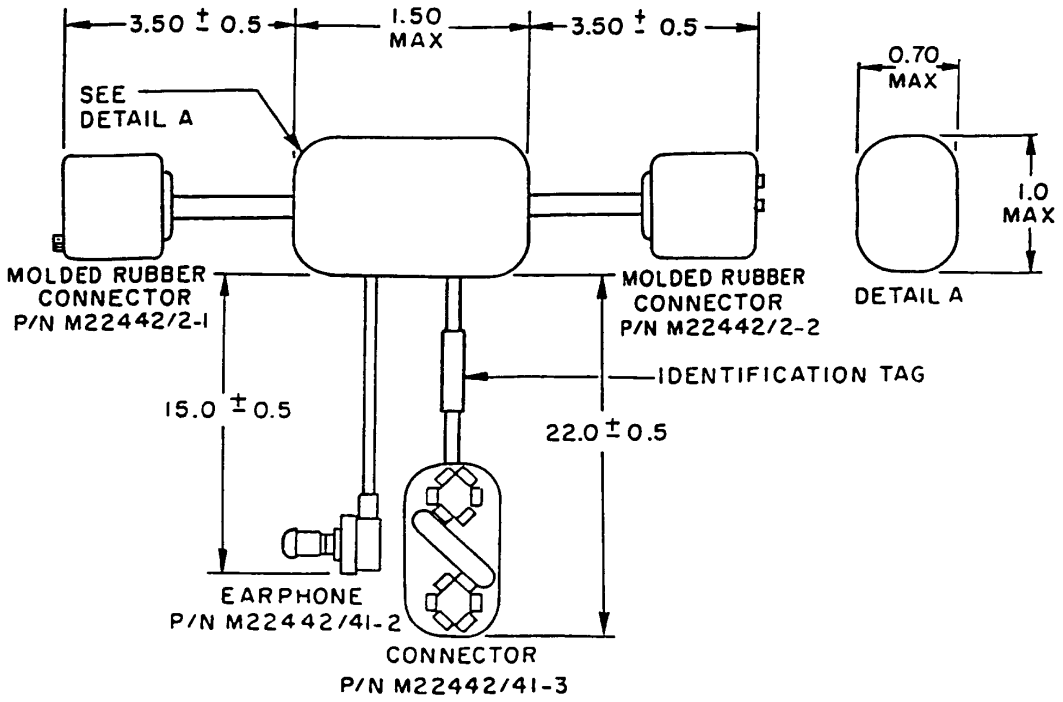


FIGURE 2. Cable assembly P/N M22442/41-4.

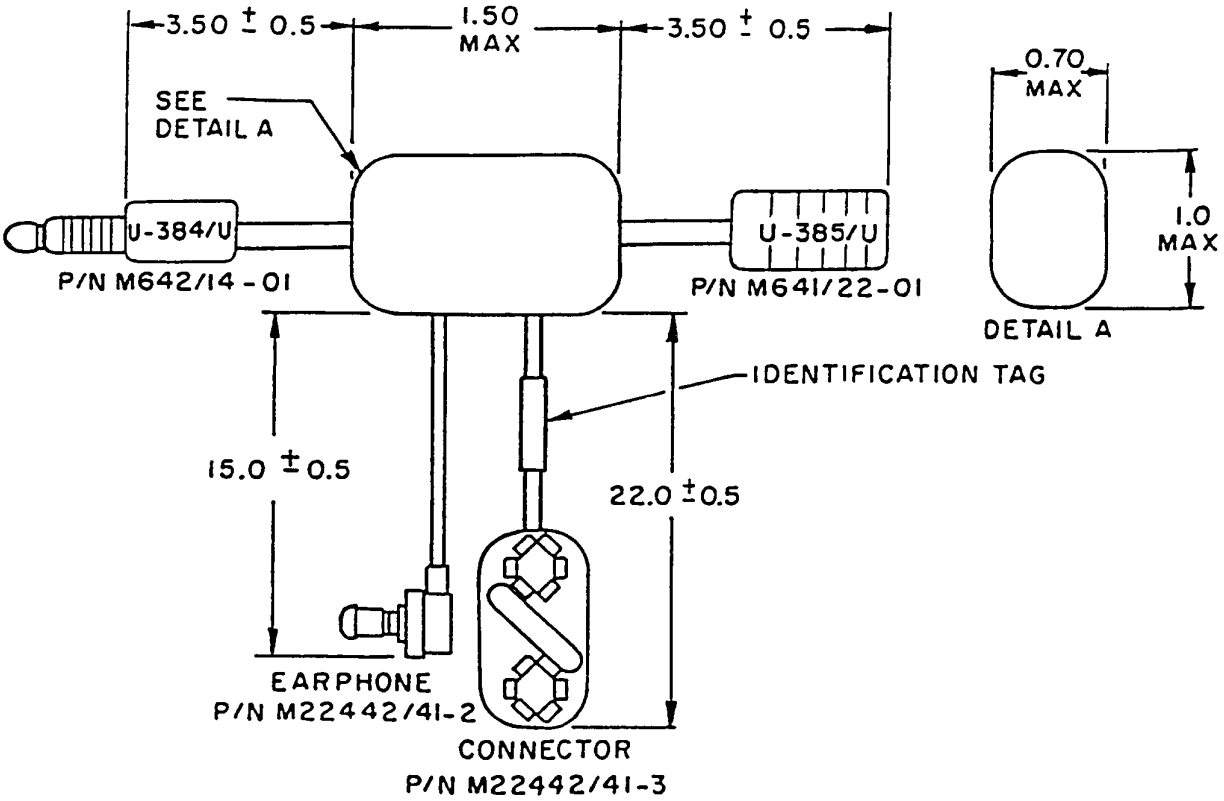


FIGURE 3. Cable assembly P/N M22442/41-5.

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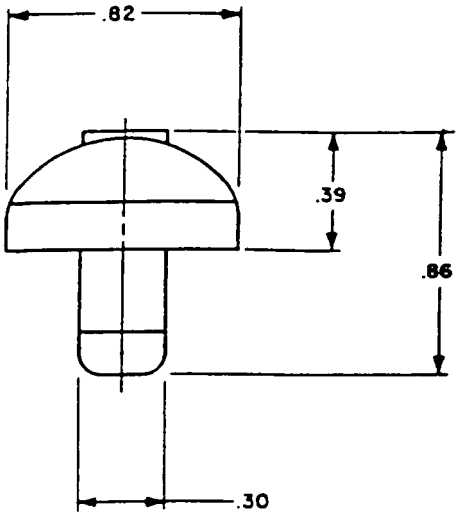


FIGURE 4. Earphone P/N M22442/41-2.

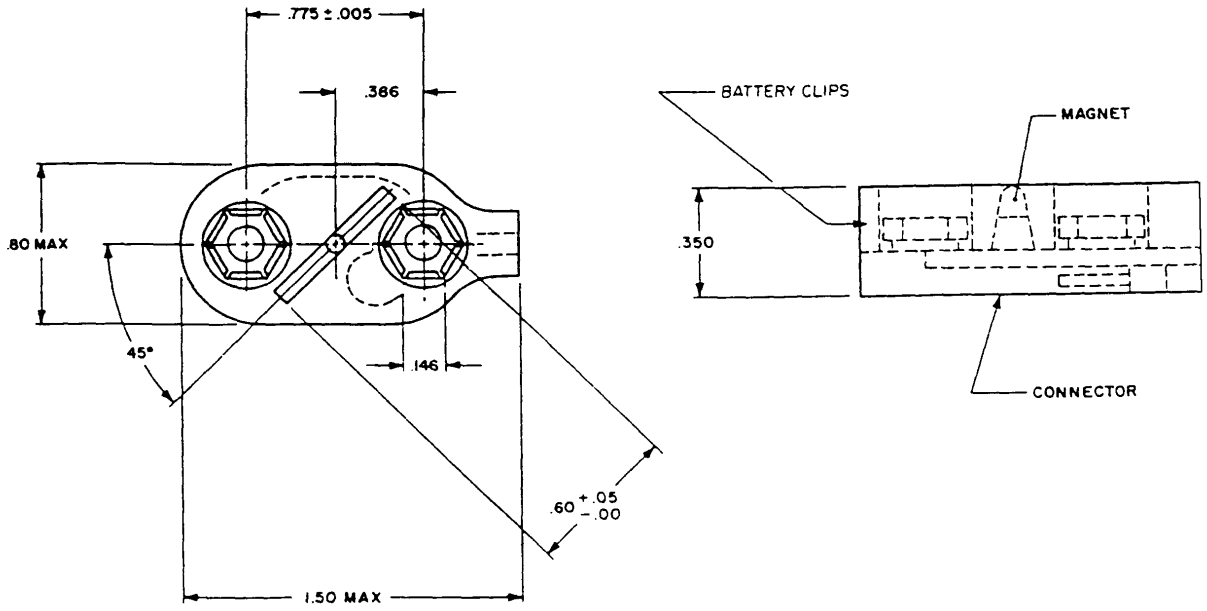
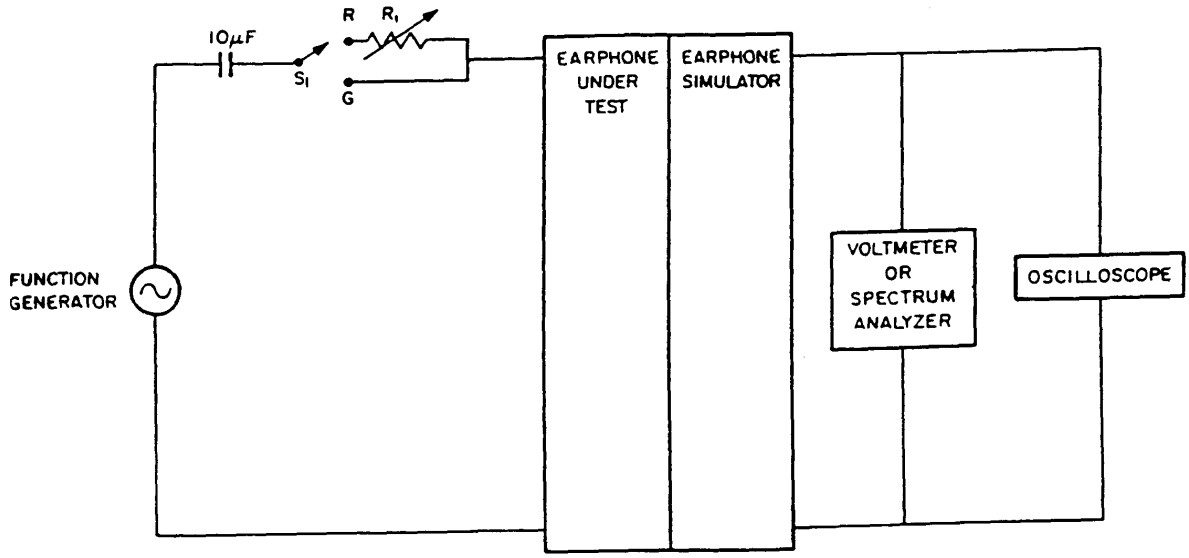


FIGURE 5. Survival radio connector P/N M22442/41-3.

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NOTES:

1. An oscilloscope is required to ascertain if any wave shape distortion or oscillation is present while performing tests.

2. R_1 is a variable resistor (or resistance decode box) which can be adjusted to any value between the limits of 4000 ohms ± 20 percent. The accuracy shall be .5 percent or better.

3. Switch S_1 shall be placed in the "R" position when performing the earphone impedance test, and placed in the "G" position when performing other tests.

4. The ear simulator shall meet the requirements of ASA Standard S3.25/1979. A possible source of supply is : Bruel and Kjaer ear simulator type 4157. The ear simulator should consist of the following parts as a minimum: microphone, microphone preamplifier, power supply.

FIGURE 6. Earphone test circuit.

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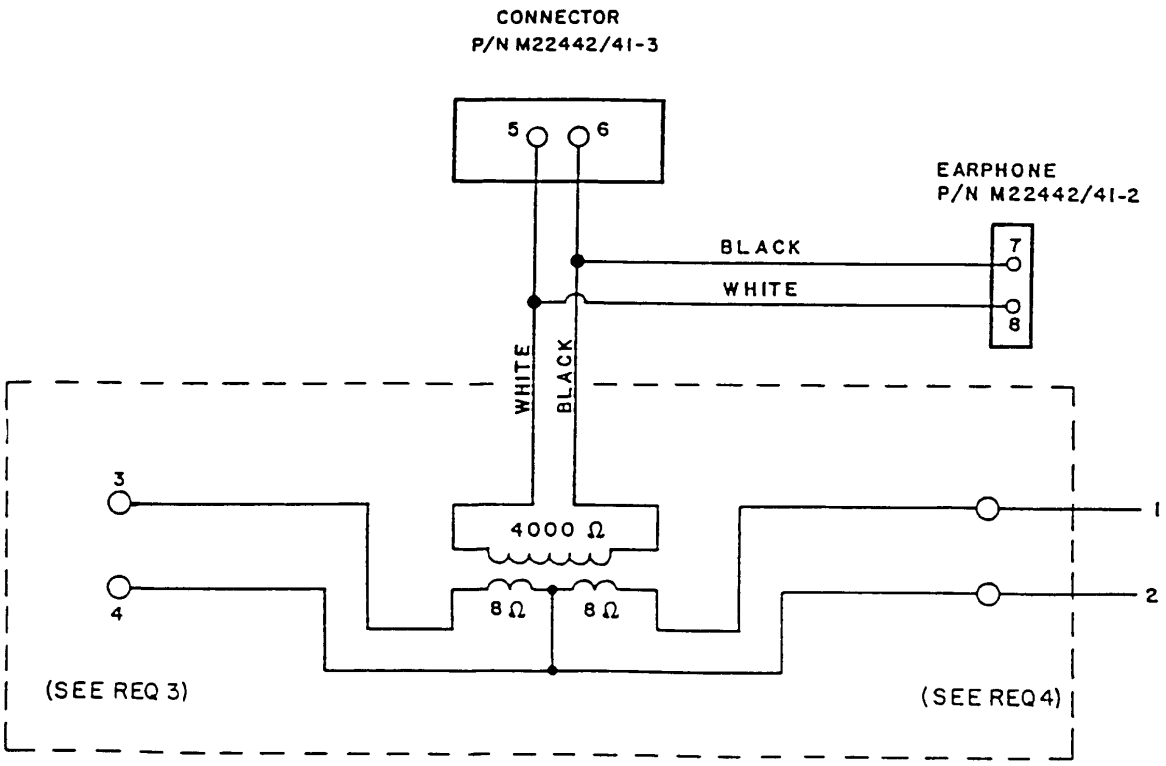


FIGURE 7. Cable assembly schematic P/N M22442/41.

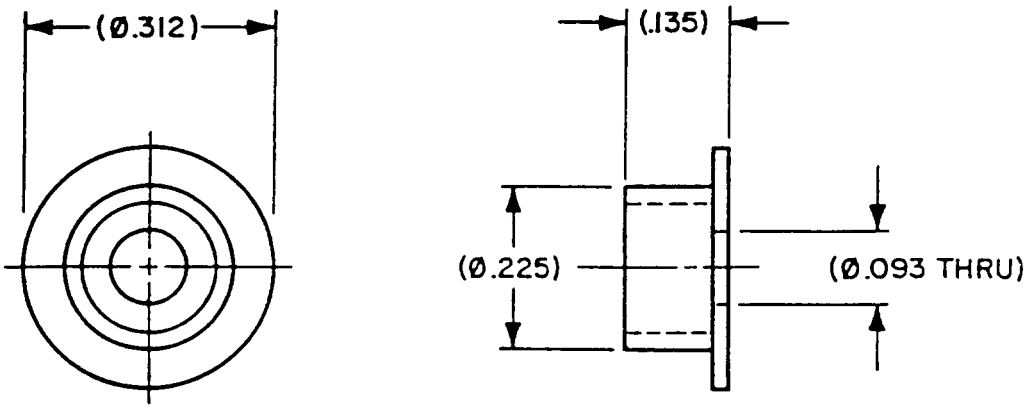


FIGURE 8. Test plugs for M22442/41-3 connector.

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APPLICABLE DOCUMENTS:

MIL-J-641	-	Jack, Telephone, General Specification for
MIL-P-642	-	Plugs, Telephone and Accessory Screws, General Specification for
MIL-C-l0392	-	Cord, Electrical (Audio, Miniature)(Inactive for New Design)
MIL-C-55668		Cord, Electrical, Audio Subminiature (Retractile and Straight)

DRAWINGS:

AIR FORCE

57B12662	-	Connector, Plug, Electrical U-l73/U Assembly of
57Bl2664	-	Tip, Electrical Plug Connector

Non-Government Publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-B16	-	Rod Brass, Free-Cutting, Bar and Shapes for use in Screw Machines (DoD adopted)
ASTM-B194	-	Copper-Beryllium Alloy Plate, Sheet, Strip, and Rolled Bar (DoD adopted)

(Application for copies should be addressed to the ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959 or webmastr@astm.org).

ACOUSTICAL SOCIETY OF AMERICA (ASA)

ASA-S3.25/1979	Occluded Ear Simulator

(Application for copies should be addressed to Customer Service American Institute of Physics Suite 1NO1, 2 Huntington Quadrangle Melville, NY 11747-4502 or asa.aip.org)

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REQUIREMENTS:

1. The cable (cord) shall be in accordance with the requirements of MIL-C-10392 Type WM-59A/U, or MIL-C-55668 Type III 2 conductor, or equal.

2. The junction block shall contain a jack capable of mating with Air Force Drawings 57B12662 and 57B12664 which govern the requirements of connector U-l73/U.

3. The junction block shall contain two pins made in accordance with Air Force Drawing 57B12664. The pins shall be arranged to form a U-173/U type connector with the dimensions specified in Air Force 57B12662.

4. The junction block assembly shall contain an impedance matching transformer made in accordance with M22442/l-8.

5. Sampling and testing procedures for the plug and jack shall be in accordance with MIL-P-642 and MIL-J-64l, respectively.

6. The plug and jack in the junction block and the M22442/4l-3 connector shall pass the following tests when tested in accordance with MIL-P-642 (see 3.4.4) and MIL-J-64l (see 3.4.5), respectively.

   a. Visual and mechanical.

   b. Dielectric strength.

   c. Contact resistance.

   d. Insertion and withdrawal force.

   e. Spring life.

   f. Salt spray.

   g. Insulation resistance.

   h. Moisture resistance.

7. When the plug and jack are mated with their proper connector, the insertion force shall be 8 pounds maximum. The withdrawal force shall be 6 ± 2 pounds.

8. After the spring life test, the withdrawal force shall be greater than 60 percent of the withdrawal force measured prior to test.

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9. Metal parts for the plug and jack shall meet all the requirements and tests of MIL-P-642 (see 3.4.4) and MIL-J-64l (see 3.4.5), respectively, except as indicated on Air Force Drawing 57Bl2664.

10. The M22442/4l-l adapter shall contain an earphone assembly which shall meet the applicable dimensions and operational requirements of MIL-E-25670 unless indicated in this specification sheet.

11. The earphone dimensions shall be in accordance with Figure 4.

12. The color of the earphone shall be black.

13. Earphone performance:

a. When tested in accordance with Figure 6 herein, the frequency response of the earphone shall not vary by more than ±3 dB from the 1000 Hz value when the input frequency is varied from 1000 Hz to 3000 Hz with a nominal input signal of 1.6 volts RMS. The frequency response may vary ±6 dB from the 1000 Hz value when the input frequency is varied from 300 Hz to 1000 Hz and from 3000 Hz to 6000 Hz.

b. Impedance of the earphone at 1000 Hz shall be 4000 ohms ±10 percent when tested in accordance with Figure 6 herein, with a nominal input signal of 1.6 volts RMS. The impedance over the frequency range of 300 to 3000 Hz shall be within 10 percent of the impedance recorded at 1000 Hz.

c. The RMS harmonic distortion in the acoustic output of the earphone shall not exceed 20 percent at any fundamental frequency between 300 and 3000 Hz when tested using the circuit shown on Figure 6 of this specification sheet, with an input signal of 2.12 volts RMS.

d. The earphone shall operate with an input signal of 2.12 volts RMS.

14. The material used for the earphone shall be as follows:

a. Earphone: Cellulose acetate butyrate (CAB) or equivalent (as approved by the qualifying activity). The color shall be black.

b. Earphone face: black anodized aluminum or equivalent (as approved by the qualifying activity).

c. Earphone holder: Half hard brass, alloy 360, in accordance with ASTM-B16 or equivalent (as approved by the qualifying activity). The earphone holder shall be nickel plated .0005 to .001 inches. The earphone holder shall have a sleeve made of vinyl plastic or equivalent (as approved by the qualifying activity), and shall be capable of being inserted comfortably into the ear.

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d. Earphone cable cord: In accordance with requirement 1. The earphone shall be molded to the cable assembly and shall be waterproof. The cable shall not separate from the earphone assembly when subjected to a 10 pound pull in any direction.

15. M22442/4l-3 connector:

a. The connector shall be made of semi-rigid vinyl plastic or equivalent (as approved by the qualifying activity). The color shall be black.

b. Metal parts of the connector shall be made of beryllium copper, alloy 172, temper AT, in accordance with ASTM-B194 or equivalent (as approved by the qualifying activity). The plugs shall be nickel plated .0005 to .001 inches.

c. The connector shall mate securely to the test plugs shown on Figure 8. The connector shall be unmated and then securely remated to the test plugs.

d. When mated with the proper connector of the survival radio, the mating force shall be 3 ±1 pounds. The unmating force shall be l0 ± l pound.

e. The connector shall contain a magnet and battery clips in accordance with Figure 5.

f. The magnet shall be made from a rod of cast Alnico 5 ground finish, or equivalent (as approved by the qualifying activity), and shall produce 1200 ± 100 Gauss when measured at the end of the rod using a Hall Effect gaussmeter.

16. When tested in accordance with Figure 7, the DC resistance shall be as indicated when tested with a suitable ohmmeter.

a. Contact l and contact 2, less than 10 ohms.

b. Contact 3 and contact 4, less than l0 ohms.

c. Contact 5 and contact 6, less than 150 ohms.

l7. The cable assembly shall not be damaged by immersion in sea water (8 percent salt by weight) at a depth of 3 meters for l0 minutes. The cable assembly shall meet all requirements immediately after immersion without any drying effort other than shaking loose water from the surface.

l8. The cable assembly shall meet the requirements specified herein and be tested in accordance with the tests listed in Table I.

NOTES:

l. This cable assembly was designed to meet the requirements of the AN/PRC-90 and AN/PRC-112(V) survival radios. It may be used on any survival radio if applicable.

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2. All dimensions are in inches.

3. Tolerances are molded parts ± .05, metal parts ± .005 inches unless otherwise specified.

CHANGES FROM PREVIOUS ISSUE: The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

TABLE I. Tests

Inspection	Requirement Paragraph	Test Paragraph	Group I Qualification Tests	Group II Quality Conformance Test 100%	Group III Sample Testing Per 4.4.1
GROUP A (ALL SAMPLE UNITS) 1/					
Visual & Mechanical Inspection	3.1 & 3.4	4.5.2	X	X	
Soldering	3.4.18	4.5.2.1	X	X	
Transformer Efficiency 2/	5	10a	X		X
Transformer Input Matching Impedance 2/	6	10b	X		X
Transformer Distortion 2/	7	10c	X		X
Transformer dc Winding Resistance 2/	8	10d	X		X
Transformer Output 2/	9	10e	X	X	
Continuity	3.5.1	4.5.3	X	X	

1/ Minimum tests for QPL retention (see paragraph 4.3 3)
2/ The number indicates the requirement or test specified in MIL-DTL-22442/1

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TABLE I. Tests - Continued.

Inspection	Requirement Paragraph	Test Paragraph	Group I Qualification Tests	Group II Quality Conformance Test 100%	Group III Sample Testing Per 4.4.1
GROUP B					
Dielectric Withstanding Voltage	3.5.2	4.5.4	X		X
Gauging 1/	3.5.3	4.5.5	X		X
Pull and Twist Test 1/	3.5.4	4.5.6	X		X
Durometer Hardness	3.5.10	4.5.12	X		X
Bend Relief Flexibility	3.5.5	4.5.7	X		X
GROUP C					
Low Temperature Flexibility	3.5.6	4.5.8	X		X
Durability 1/	3.5.7	4.5.9	X		X
Ozone Resistance 1/	3.5.8	4.5.10	X		X
Jacket Elongation	3.5.9	4.5.11	X		X
GROUP D 1/					
Salt Fog	3.6.2	4.6.2	X		
Vibration	3.6.4	4.6.4	X		
Humidity	3.6.5	4.6.5	X		
Thermal Shock	3.6.5	4.6.6	X		

1/ Minimum tests for QPL retention (see paragraph 4.3.3)

Preparing activity:
Navy - AS
(Project No. 5995-0l44)

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